Lecture 42

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1 Mathematical Paradoxes

1.1 The Horse Paradox (Fake Induction)

Theorem: All horses have the same color. **Proof (by induction):**

- **Base case ($n = 1$):** In a set containing only one horse, clearly all horses have the same color.
- **Inductive step:** Assume the statement is true for any set of n horses. Consider a set of $n + 1$ horses $\{h_1, h_2, \ldots, h_n, h_{n+1}\}$.
  1. Remove h_{n+1}. The remaining set $\{h_1, \ldots, h_n\}$ has n horses, so by hypothesis, they share the same color.
  2. Remove h_1. The remaining set $\{h_2, \ldots, h_{n+1}\}$ has n horses, so they share the same color.
  3. Therefore, h_1 has the same color as the middle horses, and h_{n+1} has the same color as the middle horses. Thus, all $n + 1$ horses have the same color.

The Flaw: The argument fails at $n = 2$ (going from 1 to 2). The two subsets $\{h_1\}$ and $\{h_2\}$ have no intersection (no "middle horses") to carry the color property across.

1.2 Set Theory Paradoxes

1.2.1 Russell's Paradox

Let $A = \{x : x \notin x\}$. **Question:** Is $A \in A$?

- If $A \in A$, then by definition $A \notin A$ (contradiction).
- If $A \notin A$, then by definition $A \in A$ (contradiction).

1.2.2 The Barber Paradox

A male barber shaves everyone who does not shave himself. **Question:** Does the barber shave himself?

- If he does, he shouldn't.
- If he doesn't, he must.

1.3 Other Paradoxes

1.3.1 Berry's Paradox

Consider the set of natural numbers. Define n as: "The smallest positive integer not definable in under sixty letters." The definition itself has fewer than sixty letters, leading to a contradiction.

1.3.2 The Liar Paradox

Epimenides, a Cretan, stated: "All Cretans are liars."

- If true, he is lying (contradiction).
- If false, then there exists a Cretan who tells the truth (consistent, provided Epimenides is lying).

1.3.3 Interesting Numbers

Theorem: All natural numbers are interesting. **Proof (by contradiction):** Suppose there are non-interesting numbers. Let S be the set of non-interesting numbers. Since $S \subset \mathbb{N}$, by the Well-Ordering Principle, S has a minimum element $s = \min(S)$. However, s is the "smallest non-interesting number," which makes it a very interesting property. Thus s is interesting, creating a contradiction.

1.4 Axiom of Choice

Let $(S_i)_{i \in I}$ be an indexed family of nonempty sets. The Axiom of Choice states that there exists a choice function (Cartesian product is non-empty):

$$\prod_{i \in I} S_i \neq \emptyset$$

This axiom leads to counter-intuitive results like the **Banach-Tarski Paradox**.

2 Logic Theorems

2.1 The Drinker Principle

Theorem: In every pub, there is a person such that if this person drinks, then everyone is drinking.

$$\exists x (D(x) \implies \forall y D(y))$$

Proof:

1. **Case 1:** Everyone drinks. Then the statement is true for any person.
2. **Case 2:** There is a person who doesn't drink. Let this person be P. The premise "P drinks" is false. In logic, False $\implies$ Anything is a True statement.

3 Constructive vs. Non-Constructive Proofs

3.1 Problem

Prove there exist two irrational numbers x and y such that $x^y \in \mathbb{Q}$.

3.2 Non-Constructive Solution

Consider the number $\sqrt{2}^{\sqrt{2}}$.

1. Case 1: If $\sqrt{2}^{\sqrt{2}} \in \mathbb{Q}$, let $x = \sqrt{2}$ and $y = \sqrt{2}$. Done.
2. Case 2: If $\sqrt{2}^{\sqrt{2}} \notin \mathbb{Q}$, let $x = \sqrt{2}^{\sqrt{2}}$ and $y = \sqrt{2}$. Then:

$$x^y = (\sqrt{2}^{\sqrt{2}})^{\sqrt{2}} = \sqrt{2}^{\sqrt{2}\cdot\sqrt{2}} = \sqrt{2}^2 = 2 \in \mathbb{Q}$$

We proved existence without knowing which case is true.

3.3 Constructive Solution

Let $x = \sqrt{2}$ and $y = \log_{\sqrt{2}} 3$. Then $x^y = (\sqrt{2})^{\log_{\sqrt{2}} 3} = 3 \in \mathbb{Q}$. **Proof that y is irrational:** Suppose $y = \frac{m}{n} \in \mathbb{Q}$.

$$\log_{\sqrt{2}} 3 = \frac{m}{n} \iff (\sqrt{2})^{\frac{m}{n}} = 3$$

$$\sqrt{2}^m = 3^n \implies 2^{m/2} = 3^n \implies 2^m = 3^{2n}$$

By the Fundamental Theorem of Arithmetic, 2^m (even factors) cannot equal 3^{2n} (odd factors) unless $m = n = 0$, which is impossible here. Thus $y \notin \mathbb{Q}$.

4 Famous Theorems and Series

4.1 Fermat's Last Theorem

The equation $x^n + y^n = z^n$ has no integer solutions for $n \geq 3$.

- For $n = 2$ (Pythagoras): $3^2 + 4^2 = 5^2$.
- Proven completely by Andrew Wiles in 1995.

4.2 The Harmonic Series

Consider the sequence of partial sums:

$$a_n = \sum_{k=1}^{n} \frac{1}{k} = 1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n}$$

4.2.1 Divergence Proof (Oresme)

We group terms to show the sum grows without bound. Consider terms up to 2^n:

$$S = 1 + \frac{1}{2} + \left(\frac{1}{3} + \frac{1}{4}\right) + \left(\frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8}\right) + \dots$$

$$S > 1 + \frac{1}{2} + \left(\frac{1}{4} + \frac{1}{4}\right) + \left(\frac{1}{8} + \cdots + \frac{1}{8}\right) + \dots$$

$$S \geq 1 + \frac{1}{2} + \frac{1}{2} + \frac{1}{2} + \dots$$

For 2^k terms, the sum is $\geq 1 + \frac{k}{2}$.

$$\lim_{n \to \infty} a_n = +\infty$$

4.2.2 Euler-Mascheroni Constant

The harmonic series diverges logarithmically:

$$\sum_{k=1}^{n} \frac{1}{k} \sim \ln n + \gamma$$

where $\gamma \approx 0.57$ is the Euler-Mascheroni constant.

5 Iterative Methods

5.1 Square Root Approximation (Heron's Method)

To find $\sqrt{q}$: Let $x_1 = 1$. Define the recurrence:

$$x_{n+1} = \frac{1}{2}\left(x_n + \frac{q}{x_n}\right)$$

Then:

$$\lim_{n\to\infty} x_n = \sqrt{q}$$